

Collateral Constraints and Asset Composition*

Chengzi Yi

National Bank of Slovakia

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Abstract

This paper studies how collateral constraints shape firms' asset composition between real estate and non-real-estate capital. I build a dynamic two-asset model with asset-specific convex and non-convex adjustment costs, as well as asset-specific pledgeability. I estimate the model using data on Chinese listed manufacturing firms. The estimates indicate greater pledgeability of real estate but substantially higher fixed costs of adjusting it. With adjustment and financial frictions, the model matches the observed right-skewed distribution of the real-estate-to-non-real-estate ratio and the declining share of real estate in total capital as capital size increases. Removing collateral constraints eliminates precautionary investment and reduces aggregate capital and revenue; removing both collateral constraints and fixed costs raises aggregate capital and revenue. In both counterfactuals, aggregate TFPR increases. In a real-estate-crisis scenario in which real estate becomes non-pledgeable, investment tilts towards non-real estate because of its higher collateral value, and aggregate capital rises; however, misallocation worsens: aggregate TFPR falls by 3.6% and revenue by 0.3%.

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1 Introduction

Firms allocate investment across different types of capital, including non-real-estate assets, such as equipment and machinery, and real estate assets, such as industrial structures and commercial properties. These categories of capital differ significantly in their specificity, divisibility, and liquidity. These differences have two implications. First, adjusting these capital goods incurs varying costs, influencing firms' investment portfolio decisions. Second, in a frictional economy with collateral constraints, these goods also differ in pledgeability, adding another layer of complexity to investment dynamics. This paper shows that, for Chinese manufacturing firms, real estate assets have higher pledgeability, but adjustments to them incur higher fixed costs than adjustments to non-real-estate assets. These differences help explain firms' capital composition and affect aggregate outcomes.

Studying firms' investment allocation in real estate capital becomes especially relevant in the context of the recent real estate crisis in China. The crisis, epitomised by the 2021 collapse of Evergrande, one of the country's largest property developers, highlights systemic vulnerabilities in the Chinese real estate market. The sharp decline in real estate prices, coupled with restrictive real estate policies, can limit firms' ability to pledge real estate assets for financing. This dynamic is particularly concerning for manufacturing firms that rely on real estate collateral to fund investments.

Using a sample of Chinese listed manufacturing firms, I document two salient features of asset composition. First, the ratio of real estate to non-real-estate assets (RE/Non-RE) is highly right-skewed across firms. Second, firms with smaller total capital allocate a larger share of capital to real estate than larger firms.

To rationalise these patterns, I develop a two-asset capital adjustment model. The two asset types, real estate and non-real-estate capital, are production inputs combined through a CES aggregator. Adjustments to both assets involve asset-specific convex and nonconvex costs. Investment, including adjustment costs, is financed with internal funds and external borrowing, with the latter constrained by collateral limits.

The elasticity of substitution between the two assets in production, adjustment costs, and collateral constraints jointly determine firms' investment dynamics and asset composition. Both convex adjustment costs and collateral constraints smooth the capital accumulation process, whereas nonconvex costs lead to lumpy adjustments. Collateral constraints affect the investment trade-off in two ways: first, a constraint that binds today increases the benefit of disinvestment; second, a potentially binding future constraint makes collateral more valuable, increasing the benefit of investment. Consequently, firms' asset composition may adjust in anticipation of future constraints, driven by differences in the pledgeability of real estate and non-real-estate assets.

I discipline the model with empirical moments from the data. I estimate the decreasing-returns-to-scale revenue function, the CES aggregator, and idiosyncratic profitability shocks

using the generalised method of moments, with moment conditions derived from the quasi-first-differenced revenue function. The estimates suggest that the two types of capital inputs are perfect complements in production. This implies that, in a frictionless economy, there is an optimal asset composition, and the ratio of real estate to non-real-estate assets remains constant. The empirical distribution, however, is highly skewed: its average lies above the technology-implied frictionless ratio, even though the median is close to it. Intuitively, introducing adjustment and financial frictions helps explain this non-degenerate distribution.

I then estimate the adjustment-cost and pledgeability parameters for both assets using the simulated method of moments, targeting moments of investment dynamics and capital composition. The two assets differ in two key respects. First, the fixed costs of adjusting real estate, which average 24% of flow revenue when investing, are significantly higher than for non-real-estate. Second, real estate is more pledgeable than non-real-estate: one unit of non-real-estate can secure external financing equivalent to 2.328 units of non-real-estate, whereas the corresponding figure for real estate is 2.547.

The model reproduces the untargeted negative relationship between firm size and the real-estate share. The mechanism combines two asymmetries. High fixed costs of real-estate adjustment make real-estate investments lumpy: firms often leave real estate inactive while adjusting non-real-estate capital in response to profitability shocks. With decreasing returns to scale, smaller firms have higher marginal returns to capital, so they have stronger incentives to invest in real estate despite the fixed costs. At the same time, higher real-estate pledgeability makes real estate valuable as collateral when constraints may bind, especially for smaller and more constrained firms. Comparisons with no-constraints, internal-funds-only, and symmetric-pledgeability specifications show that this combination is needed to match both the dispersion of asset composition and the relationship between asset composition and firm size.

The two key frictions, collateral constraints and fixed adjustment costs, both move asset composition away from the optimum implied by the production technology, but they have different implications for aggregate outcomes. Counterfactuals that progressively remove these frictions show that, as frictions decrease, asset composition moves closer to the technology-implied optimum. The average RE/Non-RE ratio falls by 16% when collateral constraints are removed and by 28% when both collateral constraints and fixed adjustment costs are removed. The allocation across firms improves in both scenarios, as aggregate TFPR increases. When collateral constraints are removed, profitable firms can grow larger, and aggregate TFPR rises by 5%. By contrast, when fixed adjustment costs are also removed, aggregate TFPR increases by only 3%, indicating that high fixed costs act as a selection mechanism that allows only sufficiently profitable firms to grow larger. Removing these frictions has different effects on aggregate capital and revenue. Without collateral constraints, capital no longer carries collateral value, eliminating the precautionary incentive to invest. Therefore, aggregate capital decreases by 4%, and aggregate revenue correspondingly decreases by 0.8%. If fixed costs are also removed, aggregate capital increases by 36.7% and aggregate revenue by 31.8%.

I use the estimated model to analyse a scenario in which real estate becomes entirely non-pledgeable. In a real estate crisis, real estate prices often decline and become more volatile. Both changes reduce collateral value: lower prices directly reduce the value of collateral, limiting its use for securing financing, while greater volatility lowers the lowest possible price realisation and therefore lowers pledgeable value under the incentive compatibility condition. When real estate is non-pledgeable, collateral constraints tighten and the collateral value of non-real-estate capital increases. Consequently, aggregate capital rises by 3%, and the average RE/Non-RE ratio decreases by 9%, bringing asset composition closer to the technology-implied optimum. However, despite the increase in aggregate capital and the improved asset composition within firms, the economy experiences greater distortions, as reflected in a 3.6% decline in aggregate TFPR and a 0.3% drop in aggregate revenue.

The remainder of this paper is organised as follows. Section 2 summarises the related literature. Section 3 documents the data and empirical patterns. Section 4 presents the dynamic model and discusses the mechanism. Section 5 discusses the GMM and SMM estimation procedures and the results. Section 6 presents the counterfactual exercises. Section 7 concludes.

2 Related Literature

This paper highlights the importance of real estate as collateral for financing investments. Chaney et al. (2012) and Gan (2007) demonstrate that real estate assets play a critical role as collateral in the US and Japan, showing that changes in real estate values due to price fluctuations have significant impacts on firms' capital accumulation—a mechanism referred to as the “collateral channel.” Bahaj et al. (2020) provide evidence that firm owners' homes serve as a source of finance. In the context of the Chinese economy, Wu et al. (2015) find no evidence of a collateral channel effect, whereas Chen et al. (2015) document a significant collateral channel effect for private firms. These studies leverage real estate price shocks and exogenous variation in real estate values to examine the effect of collateral value on firm investment.

This paper contributes to the literature by investigating the collateral channel effect in the Chinese market using a structural approach. It provides a framework for re-examining empirical evidence on the role of collateral in China and other markets. The closest related study is Catherine et al. (2022), which estimates a dynamic investment model with collateral constraints and integrates it into a general equilibrium framework to quantify the aggregate effects of financing constraints. However, their model assumes exogenous and constant real estate holdings across all firms and periods. This paper allows for endogenous decisions on real estate assets and directly addresses the potential endogeneity in real estate investment.

This paper also relates to the literature on financial frictions and investment composition, which examines the implications of financial frictions when firms have heterogeneous in-

vestment projects (Matsuyama, 2007; Aghion et al., 2010; Perez-Orive, 2016; Ottonello and Winberry). Theoretical studies show that firms tend to shift towards investment projects that are less subject to financial imperfections when credit constraints tighten. Empirically, Kermani and Ma (2023) provide evidence of heterogeneity in the liquidation values of different types of capital assets. In this paper, I distinguish real estate assets from other capital assets and estimate the production technology, adjustment costs, and pledgeability parameters that jointly explain firms’ investment dynamics. The findings reveal heterogeneous pledgeability, which further enables a quantification of the effects of financial frictions on asset composition.

Finally, this paper contributes to the literature on capital adjustment costs, building on work by Abel and Eberly (1994), Abel and Eberly (1999), Doms and Dunne (1998), and Cooper and Haltiwanger (2006), which highlight the role of nonconvex adjustment costs and examine the lumpiness in investment dynamics. Midrigan and Xu incorporate both nonconvex capital adjustment costs and borrowing frictions into their model to evaluate the extent to which these frictions explain resource misallocation. Yan (2012) estimates a capital adjustment model with nonconvex costs for Chinese manufacturing plants by plant ownership type and quantifies the source of misallocation across ownership types. As for different types of capital assets, Nilsen and Schiantarelli (2003) distinguishes between equipment and buildings and documents empirical evidence of more inaction and more spikes in investments in buildings. Chiavari and Goraya distinguishes between tangible and intangible assets, and their estimation shows higher costs in adjusting intangible assets. This paper complements these studies by finding a large fixed cost of adjusting real estate capital in a structural model and contributes to understanding the components of convex adjustment costs for different types of assets.

3 Data and Empirical Patterns

This section documents the salient features of capital adjustment and asset composition for Chinese listed manufacturing firms. First, investment rate distributions have substantial mass near zero with long right tails, more pronounced for real estate than for non-real-estate capital, which points to nonconvex adjustment costs. Second, the ratio of real estate to non-real-estate capital (h/k) is right-skewed across firms, and smaller firms hold a disproportionately large share of capital in real estate. Together, these patterns motivate a two-asset capital adjustment model to study the asset-specific adjustment costs and pledgeability.

3.1 Data

The sample consists of Chinese publicly listed manufacturing firms from 2003 to 2019. I draw firm-level financial data from the CSMAR database¹, which covers listed firms from 1990; the capital-goods price index is from the Chinese Statistical Bureau. Financial statements before 2003 do not separately identify real estate capital, so the main analysis sample starts in 2003. I retain firms with at least three consecutive annual observations, yielding an unbalanced panel of 2,145 firms and 19,997 firm-year observations.

Real estate capital comprises houses, buildings, and land; non-real-estate capital comprises equipment, machinery, and other facilities, following the detailed fixed-asset classification in the financial statement addendum. From the addendum I collect the original value, accumulated depreciation, and annual depreciation for total fixed assets and for real estate separately. The nominal value of non-real-estate capital is the residual. I recover capital stocks for both assets using the perpetual inventory method applied to data from 1990 to 2019; the detailed procedure is in Appendix A.1.

The investment rate is deflated net investment divided by the beginning-of-period capital stock. I drop the bottom and top percentile observations to limit measurement error from manual asset classification. Revenue is cash flow net of labour costs and before investment expenditure, measured as operating profit plus operating taxes, financing costs, depreciation, and amortisation (EBITDA). The model is specified in real terms, with all quantities in units of non-real-estate capital. I therefore deflate EBITDA by the machinery price from capital-goods price index P_k to obtain a model-consistent revenue measure.

Table 1 reports descriptive statistics for the main variables. The median firm holds less real estate than non-real-estate capital: the median RE/Non-RE ratio is 0.44, while the mean is 0.66, indicating a right-skewed distribution of asset composition. Investment rates are also skewed. The median investment rate is 0.19 for non-real-estate capital and 0.07 for real estate capital, while the corresponding means are 0.35 and 0.23. The sample consists mainly of large listed firms, with median log total assets of 21.71 and median firm age of 14 years; state-owned firms account for 41.7% of firm-year observations with nonmissing ownership information.

The full 2003–2019 sample spans several institutional and macroeconomic episodes. Appendix A.2 reports the same descriptive statistics separately for four subperiods that isolate the 2007 accounting-standard reform, the global financial crisis and the 2009–2010 four-trillion-yuan stimulus period, and the subsequent years that overlap with housing purchase restrictions during 2010–2014.² The main patterns are stable across periods: the RE/Non-

¹The balance sheet, the income statement, and the cash flow statement are taken from FS_Combas, FS_Comins, FS_Comscfi. The classification of the sector and other firm identifiers are from TRD_Co. The addendum to the financial statement (section: fixed assets) is from FN_Fn020.

²As an additional robustness sample, I also construct a 2011–2019 panel. This sample starts after the financial crisis and the main stimulus years, and it corresponds to a period with a more stable relative price

RE ratio remains right-skewed, and non-real-estate investment rates remain higher than real estate investment rates.

Table 1: Descriptive Statistics

	Mean	SD	Min	P25	Median	P75	Max	N
$\log(k)$, PIM	19.30	1.63	13.74	18.16	19.17	20.31	25.65	19,599
$\log(h)$, PIM	18.49	1.41	12.85	17.55	18.41	19.27	24.37	19,599
h/k	0.66	0.71	0.03	0.25	0.44	0.78	6.58	19,599
Investment rate: all capital	0.27	0.46	-0.31	0.04	0.13	0.33	3.99	17,548
Investment rate: k	0.35	0.59	-0.53	0.06	0.19	0.42	5.39	17,464
Investment rate: h	0.23	0.50	-0.39	0.01	0.07	0.25	4.53	17,524
$\log(\text{EBITDA}/P_k)$	19.09	1.41	9.22	18.15	18.95	19.93	24.74	18,720
$\log(\text{total assets})$	21.87	1.23	17.02	20.98	21.71	22.57	27.47	19,996
Firm age	14.68	5.93	1.00	10.00	14.00	19.00	61.00	19,997
State-owned firms (%)	41.7							19,997

Note: The table reports descriptive statistics for the 2003–2019 sample. k denotes non-real-estate capital and h denotes real estate capital, both constructed using the perpetual inventory method (PIM). The model is specified in real terms with all quantities in units of non-real-estate capital, so EBITDA is deflated by the capital-goods price index P_k and $\log(\text{EBITDA}/P_k)$ is the model-consistent revenue measure. Ownership rows report firm-year shares among nonmissing observations.

3.2 Empirical Patterns

Figure 1 plots the distributions of investment rates. The distributions have substantial mass around zero and are asymmetric, with long right tails. This pattern is particularly pronounced for real estate capital, for which investment is often close to inaction but occasionally involves large positive adjustments. These facts indicate a fixed-cost component in firms’ capital adjustment: investment does not move smoothly each year, and large adjustments occur only occasionally (Cooper and Haltiwanger, 2006). Appendix A.3 reports the investment-rate distributions after removing year or firm fixed effects. The distribution shifts to the left, but the long right tail remains. Such right-tail spikes reflect within-firm or within-year variation, ruling out aggregate shocks or persistent firm-level heterogeneity as the sole driver.

Figure 2 documents two salient features of firms’ asset composition. Panel (a) shows a right-skewed distribution of the RE/Non-RE ratio h/k : most firm-year observations have less real estate than non-real-estate capital, while a smaller group has much higher real-estate intensity. Panel (b) shows that average h/k declines with total capital stock, so smaller firms

of real-estate to non-real-estate capital.

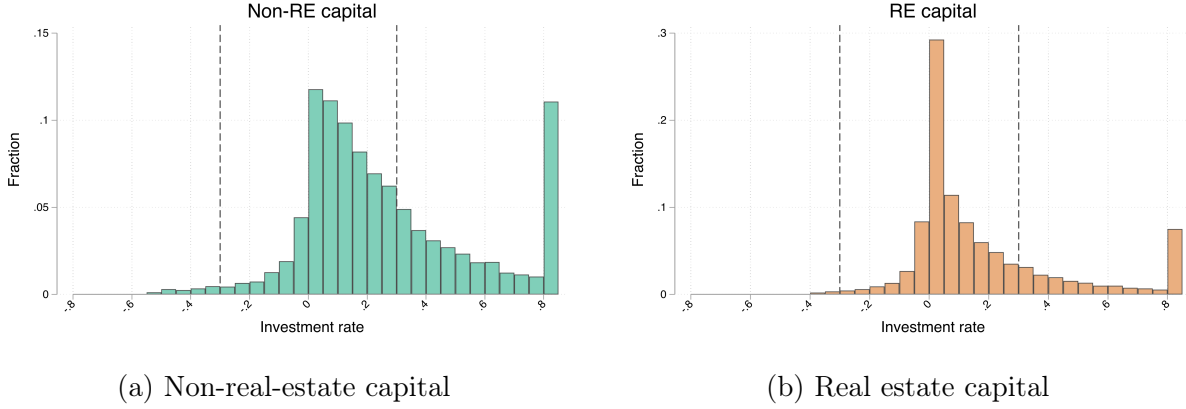


Figure 1: Distribution of Investment Rates

Note: The figure reports PIM-based investment rates in the 2003–2019 sample. Dashed lines mark the spike thresholds $\pm 30\%$. Source: Financial Reports of Listed Companies in China. Based on asset types classified by the author.

hold a disproportionately large share of capital in real estate. Appendix A.4 shows the same qualitative patterns within selected major manufacturing sectors.

Appendix A.5 also shows that the negative correlation between h/k and size remains after industry fixed effects, so the pattern is not only a sector-composition fact. Moreover, it shows that higher h/k is associated with higher loan ratios but not higher interest rates, which motivates modelling collateral as limiting the amount of borrowing rather than shifting the borrowing interest rate.

4 Capital Adjustment Model

To study firms' allocation between real estate and non-real-estate capital, I build a capital adjustment model with two heterogeneous assets. Both assets enter production, both are subject to asset-specific convex and nonconvex adjustment costs, and both can support external financing through collateral constraints. The model therefore captures how firms determine investment in the two types of capital not only as production inputs but also as costly-to-adjust and pledgeable assets.

Firms are subject to idiosyncratic profitability shocks z and choose investment in non-real-estate capital k and real estate capital h . Both assets are subject to time to build. The profitability shock follows the AR(1) process $\log(z_t) = \rho_z \log(z_{t-1}) + \sigma_z \xi_t$, with $\xi_t \sim N(0, 1)$.

The revenue function in equation 1 allows for decreasing returns to scale, with k and h bundled by a CES aggregator; σ is the elasticity of substitution between the two assets.

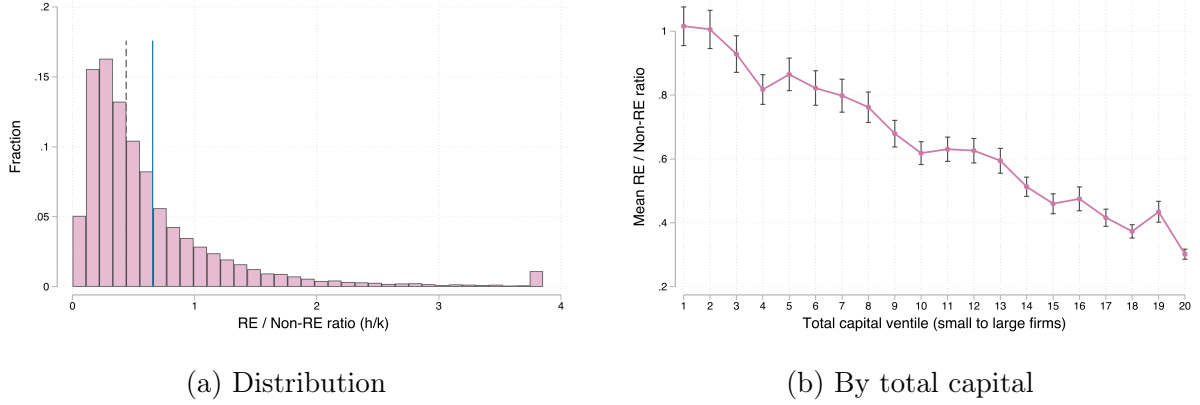


Figure 2: Distribution of RE/Non-RE ratio

Note: The figure reports the RE/Non-RE ratio in the 2003–2019 sample. Panel (a) plots the kernel density of h/k across firm-year observations; the dashed vertical line marks the mean (0.66) and the solid vertical line marks the median (0.44). Panel (b) plots mean h/k by ventiles of total capital with 95% confidence intervals. Source: Financial Reports of Listed Companies in China. Based on asset types classified by the author.

$$\pi(k, h, z) = z \{ (a^{\frac{1}{\sigma}} k^{\frac{\sigma-1}{\sigma}} + (1-a)^{\frac{1}{\sigma}} h^{\frac{\sigma-1}{\sigma}})^{\frac{\sigma}{\sigma-1}} \}^{\alpha}, \quad \alpha \leq 1. \quad (1)$$

When $\sigma \rightarrow 1$, capital input converges to the Cobb-Douglas function $(\frac{k}{a})^a (\frac{h}{1-a})^{1-a}$. When $\sigma \rightarrow 0$, capital input converges to the Leontief function $\min\{\frac{k}{a}, \frac{h}{1-a}\}$. When $\sigma \rightarrow +\infty$, capital input converges to the sum of the two assets, $k + h$.

$C(k, k')$ and $\tilde{C}(h, h')$ are the cost functions for adjusting k and h , respectively. Net investment is defined as $x_k = k' - (1 - \delta_k)k$ for non-real-estate capital and $x_h = h' - (1 - \delta_h)h$ for real estate capital. The purchase price of k is normalised to one, and p_h denotes the relative purchase price of h .

Adjustment costs include convex and nonconvex components. Convex costs smooth investment by making large adjustments more expensive at the margin, capturing the gradual building or installation process and limited financial capacity. Fixed costs generate inaction and lumpy investment, capturing adjustment episodes that require organisational restructuring, worker retraining, or similar scale-dependent costs. Asset-specific costs allow real estate and non-real-estate capital to adjust differently.

$$C(k, k') = \begin{cases} x_k + \frac{\gamma}{2} \frac{x_k^2}{k} + F_k k & \text{if } x_k \neq 0; \\ 0 & \text{if } x_k = 0; \end{cases}$$

$$\tilde{C}(h, h') = \begin{cases} p_h x_h + \frac{\omega}{2} \frac{x_h^2}{h} + F_h h & \text{if } x_h \neq 0; \\ 0 & \text{if } x_h = 0; \end{cases}$$

Fixed costs create an extensive margin, i.e., firms choose whether to adjust both assets, only k , only h , or neither asset. The value function is therefore the maximum across four discrete choices, as in equation 2. If an asset is not adjusted, its next-period stock is its depreciated current stock.

$$V(k, h, z) = \max\{V^a(k, h, z), V^k(k, h, z), V^h(k, h, z), V^n(k, h, z)\} \quad (2)$$

$V^a(k, h, z)$ is the value of adjusting both assets:

$$\begin{aligned} V^a(k, h, z) = \max_{k', h' > 0} \quad & \pi(k, h, z) - C(k, k') - \tilde{C}(h, h') + \beta \mathbb{E}V(k', h', z') \\ \text{s.t. } & C(k, k') + \tilde{C}(h, h') \leq \pi(k, h, z) + \phi_k k(1 - \delta_k) + \phi_h p_h h(1 - \delta_h) \end{aligned} \quad (3)$$

$V^k(k, h, z)$ is the value of adjusting only non-real-estate capital:

$$\begin{aligned} V^k(k, h, z) = \max_{k' > 0} \quad & \pi(k, h, z) - C(k, k') + \beta \mathbb{E}V(k', h(1 - \delta_h), z') \\ \text{s.t. } & C(k, k') \leq \pi(k, h, z) + \phi_k k(1 - \delta_k) + \phi_h p_h h(1 - \delta_h) \end{aligned} \quad (4)$$

$V^h(k, h, z)$ is the value of adjusting only real estate capital:

$$\begin{aligned} V^h(k, h, z) = \max_{h' > 0} \quad & \pi(k, h, z) - \tilde{C}(h, h') + \beta \mathbb{E}V(k(1 - \delta_k), h', z') \\ \text{s.t. } & \tilde{C}(h, h') \leq \pi(k, h, z) + \phi_k k(1 - \delta_k) + \phi_h p_h h(1 - \delta_h) \end{aligned} \quad (5)$$

$V^n(k, h, z)$ is the value of inaction:

$$V^n(k, h, z) = \pi(k, h, z) + \beta \mathbb{E}V(k(1 - \delta_k), h(1 - \delta_h), z') \quad (6)$$

Adjustment costs are financed by current revenue π and external funding. The collateral constraint limits external funding to the pledgeable value of capital, $\phi_k(1 - \delta_k)k + \phi_h p_h(1 - \delta_h)h$. The parameters ϕ_k and ϕ_h govern the pledgeability of k and h , respectively. This form, common in the literature, limits debt capacity to a fraction of pledged assets; ϕ is often interpreted as the liquidation value of capital (Kiyotaki and Moore, 1997).³

³As discussed in Kermani and Ma (2023), models with collateral constraints use a wide range of calibrated or estimated values for ϕ , from 15%-20% in Catherine et al. (2022) to 80% in Moll (2014) and Midrigan and Xu (2014). Kermani and Ma (2023) estimates the average PPE liquidation recovery rate to be 35% for a sample of U.S. firms.

In this model, the collateral constraint implies intratemporal borrowing. Firms can borrow at the beginning of the period up to the collateral value governed by ϕ_k and ϕ_h , and they repay any positive debt at the end of the period. This assumption keeps the dynamic problem tractable by abstracting from dynamic debt and cash, but it also affects the interpretation of the pledgeability estimates. If firms accumulate cash in advance of investment, as in Almeida et al. (2004), Bates et al. (2009), and James and Lirely (2022), or if debt is carried across periods, the financial constraint would be less likely to bind and the estimated pledgeability parameters would be biased upwards. Cash holdings, however, may be orthogonal to capital accumulation when driven by transaction motives, as in Miller and Orr (1968), or tax motives, as in Foley et al. (2007). Appendix B, following Görtz et al. (2023), shows that, for Chinese listed manufacturing firms, positive investment spikes are associated with an increase in debt during the adjustment period and one year after, while firms do not appear to accumulate cash before the spikes. This evidence points to limited debt rollover and limited cash hoarding, but the estimated values of ϕ_k and ϕ_h are still not directly interpretable as liquidation values. These parameters capture the pledgeability of the two assets for accessing external debt and help identify whether real estate and non-real-estate capital affect investment dynamics and asset composition beyond their roles as production inputs.

Collateral constraints affect investment through both current and future financing conditions. When the constraint binds in the current period, it raises the shadow cost of investment and makes disinvestment more attractive because selling capital relaxes the financing constraint. When the constraint may bind in the future, pledgeable capital becomes more valuable as a precautionary buffer: investment raises future borrowing capacity, while disinvestment reduces it. This forward-looking collateral motive, similar to the precautionary saving channel in Perez-Orive (2016), links the relative pledgeability of real estate and non-real-estate capital to firms' asset composition.

5 Structural Estimation

The empirical patterns discipline the quantitative model through moments of investment dynamics and asset composition. Before estimating the adjustment-cost and pledgeability parameters, I first estimate the revenue function and profitability process, which pin down the production technology and shock process used in the SMM estimation.

5.1 Estimation Procedure

The estimation proceeds in three steps. First, I estimate the revenue-function parameters and the profitability process using generalised method of moments (GMM). Second, I set the remaining externally calibrated parameters, including the discount factor, relative price of real estate capital, and depreciation rates. Third, I estimate the adjustment-cost and

pledgeability parameters by simulated method of moments (SMM), targeting moments of investment dynamics and asset composition.

Let $\Theta = [\gamma, F_k, \omega, F_h, \phi_k, \phi_h]$ collect the parameters estimated by SMM. For each candidate Θ , I solve the capital adjustment model, simulate a stationary distribution of 3000 firms for 100 periods, and compute the simulated moments $\tilde{M}(\Theta)$. The SMM estimator minimises the distance between the data moments M and the simulated moments:

$$\hat{\Theta}_{smm} = \arg \min_{\Theta} (M - \tilde{M}(\Theta))' W (M - \tilde{M}(\Theta)) \quad (7)$$

W is the weighting matrix. I obtain it by block bootstrapping the sample at the firm level. For each of the N_{sample} bootstrap samples, I compute the same moments used in the SMM objective. The weighting matrix is the inverse of $\hat{V}$, where

$$\hat{V} = \frac{1}{N_{\text{sample}}} \sum_{i=1}^{N_{\text{sample}}} (\text{moments}_{\text{sample},i} - \text{moments}_{\text{data}})' (\text{moments}_{\text{sample},i} - \text{moments}_{\text{data}}).$$

The multivariate global optimisation algorithm follows two steps similar to Arnoud et al. (2019); Appendix D.2 describes the algorithm in detail. Appendix D.3 describes the procedure for obtaining standard errors.

5.1.1 Revenue Function Estimation

The revenue-function parameters include the curvature α , the share parameter a , the elasticity of substitution σ , the persistence of the profitability process ρ_z , and the innovation standard deviation σ_z . I estimate α , a , σ , and ρ_z by GMM, using moment conditions from the quasi-first-differenced revenue function, and recover σ_z from the estimated residual process.

Equation 8 is obtained by taking logs of the revenue function and quasi-differencing, where $\mathbf{K}_{it} \equiv (a^{\frac{1}{\sigma}} k_{it}^{\frac{\sigma-1}{\sigma}} + (1-a)^{\frac{1}{\sigma}} h_{it}^{\frac{\sigma-1}{\sigma}})^{\frac{\sigma}{\sigma-1}}$. The residual ξ_{it} is the innovation to profitability at t . It is orthogonal to k_{is} , h_{is} , and π_{is} for all $s \leq t-1$. Because capital is subject to time to build, ξ_{it} is also orthogonal to contemporaneous k_{it} and h_{it} . These orthogonality conditions identify α , a , σ , and ρ_z .

$$\log \pi_{it} = \rho_z \log \pi_{it-1} + \alpha \cdot (\log \mathbf{K}_{it} - \rho_z \log \mathbf{K}_{it-1}) + \xi_{it} \quad (8)$$

I use contemporaneous k_{it} and h_{it} , first lags k_{it-1} , h_{it-1} , and π_{it-1} , and second lags k_{it-2} ,

h_{it-2} , and π_{it-2} as instruments. I also control for year effects to remove aggregate shocks⁴. When σ converges to the limits 0, 1, or $+\infty$, the aggregation function converges to Leontief, Cobb-Douglas, or linear capital aggregation, respectively. To avoid non-smoothness and numerical error near these limits, while still exploiting gradient information in optimisation, I estimate the CES functional form and compare specifications in which σ is fixed at values from 0.05 to 2. The GMM estimation follows a two-step procedure for efficiency.

The preferred specification sets $\sigma = 0.05$, suggesting that the aggregation between k and h is better approximated by a Leontief technology. Figure 3a plots the GMM objective function in the first and second stages across values of σ . The first stage uses an identity weighting matrix, while the second stage uses a feasible optimal weighting matrix. In both stages, the objective function is smallest when σ is close to zero and increases monotonically with σ . The parameter estimates are also more precise when σ is close to zero. As shown in Figure 3 panels (b), (c), and (d), the estimates jump as σ increases, and their standard errors rise. The estimates for $\hat{\rho}_z$ and $\hat{a}$ remain fairly robust when σ is sufficiently small, while $\hat{a}$ is estimated most precisely when $\sigma = 0.05$. I therefore use a CES aggregator with $\sigma = 0.05$ in the structural estimation. This smooth approximation avoids numerical errors from the Leontief limit, and simulations show that the moments are very similar when the aggregation technology is instead specified as Leontief.

After estimating α , a , σ , and ρ_z , I recover the idiosyncratic shocks $\hat{z}$ by plugging the estimates of α , a , and σ into the revenue function and removing year fixed effects. I then estimate an AR(1) process for the residualised shocks to recover the innovation variance. The innovation standard deviation σ_z is estimated at 0.5716. The persistence parameter from this AR(1) estimation is similar to the GMM estimate of ρ_z .

The parameter estimates are summarised in Table 2, and Table A3 presents the complete estimation results. The share parameter a is estimated at 0.6857. Together with the Leontief functional form, this estimate pins down the frictionless ratio between h and k , $\frac{1-\hat{a}}{\hat{a}} \approx 0.4584$. This value is close to the sample median of the empirical h/k distribution, which is about 0.4407.

The GMM estimation uses an unbalanced panel. Endogenous entry into and exit from the sample may bias the estimates, since firms enter when they go public and exit when they delist. Modelling the listing process is beyond the scope of this paper. As a robustness exercise, I apply the same GMM procedure to a strictly balanced panel. The results in Table A4 again indicate that the objective function is smallest when $\sigma = 0.05$, and the estimate of a is about 0.7. In the balanced panel, the persistence of idiosyncratic shocks is higher and the revenue curvature is lower, consistent with selection from retaining firms that survive throughout the sample period.

⁴The year fixed effects (denoted as γ_t) are included as parameters to be estimated in the GMM, and the dummies are themselves instruments, since $\mathbb{E}(\gamma_t \xi_{it}) = 0$ holds for all t .

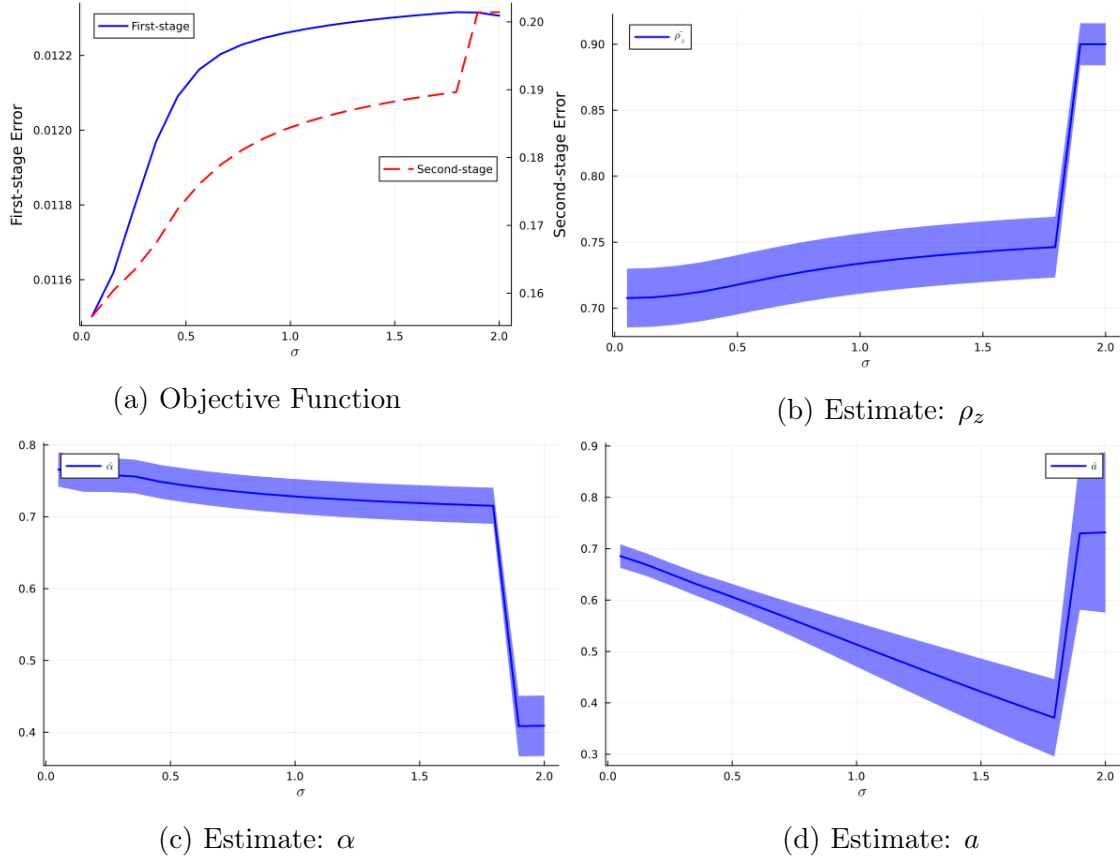


Figure 3: Revenue Function Estimation

Note: The GMM is performed to estimate ρ_z , α , and a while fixing σ at different values. Panel(a) displays the value of the objective functions obtained in the first and second stages. Panels (b), (c), (d) present the parameter estimates as functions of σ , with 95% confidence intervals included.

5.1.2 Other Pre-defined Parameters

Table 2 reports the pre-defined parameters. The discount factor is set to 0.9479, corresponding to an interest rate of 5.5%, the weighted average of the one-year benchmark RMB loan rate during the sample period. The relative price of h is 1.5522, obtained as the sample average of the estimated relative price across firm-year observations; Appendix A.6 provides details. The depreciation rates for non-real-estate and real estate capital are set to 0.1275 and 0.0487, respectively, based on the sample averages of in-use depreciation rates. These depreciation rates are also used to construct the capital stocks.

Table 2: Model Parameters

	Value	Description	Source
β	0.9479	discount factor	$\frac{1}{1+r}$, $r = 0.055$
p	1.5522	relative price of h	sample mean
δ_k	0.1275	depreciation rate of k	in-use depreciation rate
δ_h	0.0487	depreciation rate of h	in-use depreciation rate
α	0.7659	curvature of revenue function	GMM estimation
σ	0.0500	CES elasticity of substitution	Leontief aggregator
a	0.6857	share of k	GMM estimation
ρ_z	0.7077	idiosync. prof.: persistence	GMM estimation
σ_z	0.5716	idiosync. prof.: std. dev.	GMM estimation

5.1.3 Data Moments

Table 3 summarises the targeted moments used to estimate the adjustment-cost and pledgability parameters. The targeted moments include the serial correlations of the investment rates of k and h and the probabilities of positive investment spikes, defined as investment rates exceeding 30%.⁵ The targeted moments also include the sample mean and median of the h/k (RE/Non-RE) ratio. These moments discipline investment persistence, investment lumpiness, the average real estate share in capital, and the skewness of the capital-composition distribution.

I adjust the data moments for year fixed effects⁶. All moments are calculated using the unbalanced panel. Only 514 firms are listed in all years of the sample period and have

⁵Because average investment rates are high in the sample, an absolute spike threshold may not fully capture investment lumpiness. As a robustness check, Appendix E.1 reports untargeted relative investment spikes, following Power (1998), defined as investment rates that exceed normal investment rates by a large margin, namely more than 2.5 times a firm's average investment rate over the sample period.

⁶I adjust the constructed variables by regressing them on year fixed effects, obtaining the residuals, and adding back the sample averages. This filters out variation that is not due to idiosyncratic shocks and leads to lower serial correlation.

continuous information from 2003 to 2019, so balancing the panel may introduce selection bias. Nevertheless, the investment-rate moments are similar in the unbalanced and balanced panels.

Table 3: Targeted Data Moments

Value	Description	Definition
0.0508	serial correlation of i_k	$corr(i'_k, i_k)$
0.0281	serial correlation of i_h	$corr(i'_h, i_h)$
0.3527	absolute positive spikes of i_k	$spike_k^+ \equiv Pr(i_k > 30\%)$
0.2166	absolute positive spikes of i_h	$spike_h^+ \equiv Pr(i_h > 30\%)$
0.6549	sample average of h/k	$\overline{h/k}$
0.4407	sample median of h/k	$med(h/k)$

5.2 Estimation Results

Table 4 reports the estimated parameters and the corresponding simulated moments. The preferred specification is column (4), which allows collateral constraints and asymmetric pledgeability. This specification improves the fit to the targeted moments and, importantly, reproduces the untargeted negative relationship between the h/k ratio and firm size.

In addition to the targeted moments, Table 4 reports untargeted moments that describe the shape of the h/k ratio distribution. The moments $\overline{h/k}_{Q_1} \dots \overline{h/k}_{Q_5}$ are the average h/k ratios for firms in each capital-size quintile group. As shown in Figure 2 panel (b), the average h/k ratio decreases with capital size in the data. The moment $skew(h/k)$ denotes the skewness of the h/k ratio distribution.

Model Fit. I compare four model specifications. Column (1) assumes $\phi_k = \phi_h = \infty$, so collateral constraints never bind and there are no explicit financial frictions constraining external borrowing. Column (2) assumes $\phi_k = \phi_h = 0$, so collateral constraints are most restrictive and capital adjustments must be financed entirely through internal funds. Column (3) allows assets to be pledgeable but restricts $\phi_k = \phi_h > 0$. Column (4) further relaxes this restriction and allows ϕ_k and ϕ_h to differ.

The comparison between the two extreme cases in columns (1) and (2) shows how financial frictions affect firm investment. When collateral constraints are strict, investment is smoothed across periods, generating higher serial correlations and fewer investment spikes. The h/k distribution is less skewed, and the model does not generate a clear relationship between capital size and asset composition. By contrast, the model without explicit financial frictions generates a skewed h/k distribution. Its average h/k ratio is much higher than its median, and its skewness is close to the observed value. It also generates a downward trend in h/k as capital size increases, although the size-bin moments are not perfectly ordered.

The symmetric-pledgeability specification in column (3) produces a distance from the targeted moments similar to that in column (1), and the estimated common pledgeability parameter is significantly different from zero. Allowing asymmetric pledgeability in column (4) further improves model fit relative to the model with no binding collateral constraints in column (1), as the distance is significantly smaller⁷. More importantly, the preferred model generates a negative correlation between the h/k ratio and capital size, even though the size-bin moments are not targeted in the estimation.

Parameter Estimates. In the preferred specification in column (4), the fixed cost of adjusting real estate capital is much larger than the fixed cost of adjusting non-real-estate capital. The estimate of F_h is 0.2684, compared with $F_k = 0.0029$. The estimate of F_h is equivalent to 24% of flow revenue on average when firms make positive real estate investments. The same qualitative ranking, with F_h much larger than F_k , is robust across model specifications.

The high F_h helps explain investment spikes for both assets. Because k and h are complementary in production, lumpy adjustments in h lead to lumpy adjustments in k . The model therefore generates a positive spike rate in k despite the small estimated fixed cost for non-real-estate capital.

The high fixed cost of adjusting h also helps generate a non-degenerate distribution of asset composition by making real estate adjustment lumpy. Firms with the same capital stocks (k, h) may choose different k' in response to different profitability shocks while remaining inactive in h , and therefore choosing the same h' . Asset composition then varies across firms even when they start from the same capital state. High fixed costs for h also generate a higher h/k ratio among smaller firms, because smaller firms have higher marginal returns to capital and are more likely to invest even when investment in h requires paying fixed costs.

The model omits irreversibility terms $p_k^- < 1$ and $p_h^- < p_h$ because these parameters are not well identified by negative investment spikes. Non-real-estate capital has a high depreciation rate in the data, and the estimated F_h is large; as a result, changing p_k^- and p_h^- generates little additional variation in the simulated moments. Even without irreversibility, the model can still generate negative investment spikes in the simulation that are close to the observed untargted moments, as shown in Appendix E.1.

In column (4), the convex adjustment-cost estimates are $\gamma = 0.0061$ for non-real-estate capital and $\omega = 0.0233$ for real estate capital. As explained in Section 4, convex costs smooth investment and account for gradual adjustment due to limited financial capacity. This explains why the convex-cost estimates change when collateral constraints are specified differently. For example, the estimate of ω is significantly higher in the economy without explicit collateral constraints in column (1) than in the economy with the most restrictive

⁷The SMM estimation uses the optimal weighting matrix. Therefore, the distance measures follow χ^2 distributions. Since the model in column (1) is nested with the model in column (4), a χ^2 difference test can be performed to compare the models. The more complex model provides a significantly better fit with a p-value of 0.00064.

Table 4: Estimation Results

	(1) no cons.	(2) internal funds	(3) sym. pledge	(4) asym. pledge	Data
A. Model Parameters					
γ	0.0068 (0.0010)	0.0200 (0.0153)	0.0070 (0.0012)	0.0061 (0.0097)	
F_k	0.0015 (0.0015)	0.0021 (0.0006)	0.0013 (0.0005)	0.0029 (0.0048)	
ω	0.0992 (0.0278)	0.0144 (0.0094)	0.0419 (0.0087)	0.0233 (0.0925)	
F_h	0.2541 (0.0064)	0.3983 (0.0179)	0.3049 (0.0078)	0.2684 (0.0425)	
ϕ_k	$+\infty$	0	2.6737 (0.2053)	2.3283 (1.3100)	
ϕ_h	$+\infty$	0	2.6737 (0.2053)	2.5466 (0.3221)	
B. Moments					
$corr(i'_k, i_k)$	0.0080	0.0832	0.0180	0.0204	0.0508
$corr(i'_h, i_h)$	0.0293	0.0661	0.0502	0.0269	0.0281
$spike_k^+$	0.2393	0.1748	0.2441	0.2551	0.3527
$spike_h^+$	0.1420	0.0978	0.1406	0.1589	0.2166
$\overline{h/k}$	0.6951	0.5558	0.6826	0.6603	0.6549
$med(h/k)$	0.4822	0.4979	0.4754	0.4728	0.4407
$\overline{h/k}_{Q1}$	1.20	0.55	1.10	1.20	0.94
$\overline{h/k}_{Q2}$	0.65	0.56	0.65	0.62	0.80
$\overline{h/k}_{Q3}$	0.67	0.58	0.66	0.51	0.64
$\overline{h/k}_{Q4}$	0.47	0.58	0.47	0.48	0.51
$\overline{h/k}_{Q5}$	0.48	0.51	0.52	0.47	0.38
$skew(h/k)$	3.23	1.84	3.45	3.69	3.31
Distance	50.795	419.656	53.092	36.088	-

Note: The table presents the SMM estimation results for different collateral-constraint specifications. Column (1) is the no-constraints specification, column (2) allows only internal funds, column (3) imposes symmetric pledgeability, and column (4) allows asymmetric pledgeability. Panel A shows the model parameter estimates, with standard errors in parentheses, while Panel B displays the corresponding targeted and untargeted moments. The parameters in bold are estimated by targeting the moments in bold. The last column shows the data moments. “Distance” measures the discrepancy with respect to the targeted moments, as defined in equation 7.

collateral constraints in column (2). Although column (1) does not explicitly model financial frictions, convex costs can account for some gradual adjustment induced by limited financial capacity. The preferred specification, with two capital inputs and collateral constraints, helps separate convex adjustment costs from the investment smoothing generated by financial constraints.

The preferred specification estimates pledgeability parameters of $\phi_k = 2.3283$ for non-real-estate capital and $\phi_h = 2.5466$ for real estate capital. These estimates imply that both assets substantially relax firms' external financing constraints, with real estate capital providing slightly greater borrowing capacity. The magnitudes exceed one and are higher than values commonly used in macroeconomic calibrations, so they should not be read as direct estimates of liquidation values or haircuts in collateralised borrowing contracts. As discussed in Section 4, the model abstracts from dynamic cash and debt choices and from unsecured or earnings-based borrowing. The estimates therefore capture the broader external funding capacity associated with holding each type of capital, including financing channels that may be correlated with asset holdings even when they are not explicitly collateralised.

Role of Collateral Constraints. The preferred specification shows that occasionally binding collateral constraints and asymmetric pledgeability are important for matching the asset-composition pattern. In Table 4, specifications that either impose extreme collateral constraints or restrict pledgeability to be symmetric do not reproduce the negative relationship between the h/k ratio and capital size. By contrast, the specification in column (4), which allows ϕ_k and ϕ_h to differ, matches this untargeted size gradient more closely. Figure A7 shows a similar comparison in counterfactual simulations that fix γ , F_k , ω , and F_h at their preferred estimates while varying the pledgeability parameters.

Firms tend to tilt their investments towards more pledgeable assets if there are potentially binding constraints in the future. This precautionary investment motive is stronger for smaller firms, which are more likely to be constrained in the preferred simulation. Figure A8 shows that the share of financially constrained firms declines with capital size, coinciding with the decreasing h/k ratio across size groups.

5.3 Identification of Parameters

Table 5 reports elasticity matrices of the simulated moments with respect to the model parameters in the specification where $\phi_k = \phi_h = +\infty$ and in the preferred specification where $\phi_k \neq \phi_h > 0$. Each entry is the elasticity of a simulated moment with respect to a model parameter, evaluated in a neighbourhood of the parameter estimates. These elasticities indicate which moments identify which parameters. A moment that is more sensitive to a parameter is more informative about that parameter. Appendix D.3 provides further details on the computation of the elasticity matrices.

Comparing the two elasticity matrices highlights the relationship between convex adjustment

costs and financial constraints. When there are no explicit financial constraints, the serial correlations are sensitive to γ and ω . Once collateral constraints are introduced, the serial correlations become more sensitive to ϕ_k and ϕ_h and less sensitive to γ and ω .

In both model specifications, the average h/k ratio is sensitive to changes in F_h , while the median h/k ratio is insensitive, given that the elasticities are approximated around a narrow neighbourhood of the parameter estimates. In the model with collateral constraints, the sample average and median h/k ratios are sensitive to F_h , ϕ_k , and ϕ_h , consistent with the mechanisms discussed above.

Table 5: Elasticity of Moments to Parameters

a. No constraints						
Targeted Moments	Model Parameters					
	γ	F_k	ω	F_h		
$corr(i'_k, i_k)$	5.12	-2.09	13.38	-5.81		
$corr(i'_h, i_h)$	0.96	-0.77	2.92	3.04		
$spike_k^+$	0.02	-0.01	0.02	-0.09		
$spike_h^+$	0.01	-0.04	0.08	-1.14		
$\overline{h/k}$	-0.14	0.02	-0.06	0.45		
$med(h/k)$	-0.07	0.00	-0.02	0.05		
b. Preferred specification						
Targeted Moments	Model Parameters					
	γ	F_k	ω	F_h	ϕ_k	ϕ_h
$corr(i'_k, i_k)$	2.70	2.08	1.92	13.04	19.63	6.66
$corr(i'_h, i_h)$	1.57	0.24	0.06	10.54	3.45	2.92
$spike_k^+$	-0.04	-0.04	-0.02	-0.44	0.04	-0.30
$spike_h^+$	-0.02	-0.04	0.03	-0.14	0.45	0.18
$\overline{h/k}$	-0.05	-0.03	-0.02	-0.29	-0.25	-0.27
$med(h/k)$	0.00	0.00	0.00	-0.15	-0.04	0.05

6 Counterfactual Exercises

6.1 Decomposition of the Effects of Frictions

In the model, the distribution of real estate and non-real-estate capital is determined by production technology, heterogeneous adjustment costs, and heterogeneous pledgeability. I decompose these channels by comparing three counterfactual scenarios: the baseline economy with both collateral constraints and fixed adjustment costs, an economy without collateral

constraints but with fixed costs, and an economy with neither collateral constraints nor fixed adjustment costs. Panel A of Table 6 reports the results.

The average h/k ratio is 0.656 in the baseline model. It decreases by 16% when collateral constraints are removed and by 28% when both collateral constraints and fixed adjustment costs are removed. In the latter case, the average h/k ratio is close to the optimal composition implied by the Leontief aggregation technology. Aggregate fixed assets, defined as the sum of real estate and non-real-estate assets, decrease when collateral constraints are removed but increase when both collateral constraints and fixed costs are removed. Financial frictions provide additional incentives to invest in collateralisable assets in the frictional economy, resulting in over-investment and distortions in asset composition. Consistent with this mechanism, aggregate capital decreases by 4% and aggregate revenue decreases by 0.8% when collateral constraints are removed. Removing fixed costs brings asset composition closer to the optimum and allows firms to grow larger. In this counterfactual economy, aggregate capital increases by 36.7% and aggregate revenue increases by 31.8%.

Aggregate TFPR, defined as profitability weighted by capital size, also increases as collateral constraints and fixed adjustment costs are removed. Removing collateral constraints raises aggregate TFPR by 5%, as firms with positive profitability shocks can grow larger. When both collateral constraints and fixed costs are removed, aggregate TFPR increases by only 3%, suggesting that high adjustment costs act as a filter that favours firms with higher profitability.

Table 6: Counterfactual Exercises

model	$\overline{h/k}$	$\overline{h/k}, \%$	agg. TFPR	agg. revenue	agg. fixed assets
<i>Panel A. Removing frictions</i>					
baseline	0.656		1	1	1
no constraints	0.549	-16%	1.050	0.992	0.960
no constraints & no fixed costs	0.471	-28%	1.032	1.318	1.367
<i>Panel B. Real estate pledgeability</i>					
$\phi_h = 0$	0.596	-9%	0.964	0.997	1.030

Note: "No constraints" removes the collateral constraints.

6.2 Change in Real Estate Pledgeability

Panel B of Table 6 illustrates the effects of a real estate crisis in which real estate assets become non-pledgeable. This exercise captures one supply-side consequence of the recent Chinese real estate crisis. Although the crisis may affect firms' production through demand-side channels, it can also limit firms' ability to pledge real estate assets as collateral because real estate prices decline and become more volatile. This channel is especially important for

firms that rely on real estate collateral to fund investment.

In the simulation, removing the collateral value of real estate assets (h) makes financial constraints more binding and increases the relative pledgeability of non-real-estate assets (k). This shift leads to a 9% decrease in the average h/k ratio, aligning asset composition more closely with the theoretical optimum determined by production technology. Tighter constraints and the greater relative collateral value of non-real-estate assets also lead to a 3% increase in aggregate capital. These changes, however, come at a cost: aggregate TFPR declines by 3.6%, and aggregate revenue decreases by 0.3%, highlighting the distortions caused by restricted access to external financing.

7 Conclusion

This paper investigates firms' asset composition and investment behaviour in the presence of collateral constraints. For Chinese listed manufacturing firms, the ratio of real estate to non-real-estate assets is highly right-skewed, and the real-estate share of capital declines with firm size. By estimating a two-asset capital adjustment model, I show that these patterns are consistent with real estate having higher fixed adjustment costs and higher pledgeability than non-real-estate capital. High fixed costs make real-estate adjustment lumpy, while higher pledgeability gives firms an incentive to hold real estate as collateral when constraints may bind.

Counterfactuals quantify the role of these frictions. Removing collateral constraints lowers the average real-estate share and raises aggregate TFPR, while removing both collateral constraints and fixed adjustment costs moves asset composition further towards the technology-implied optimum. When real estate becomes non-pledgeable, firms shift away from real estate towards non-real-estate capital, but tighter external financing lowers aggregate TFPR and revenue.

However, certain limitations warrant consideration. Because the model abstracts from dynamic debt choice, the estimates of pledgeability should be interpreted as external funding capacity within the model rather than as observed collateral haircuts. This abstraction also limits identification, since debt-related variables cannot be directly targeted in the estimation. Future research could extend the model by making debt a dynamic decision, thereby providing a more realistic characterisation of firms' problems.

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Appendix

A Data Construction and Descriptive Robustness

This section provides the data-construction details and descriptive robustness checks that support the empirical patterns in Section 3.

A.1 Construction of Capital Stock

I construct the capital stock of both real estate and non-real-estate assets using the perpetual inventory method. The procedure is as follows:

- I obtain the gross investment each year from the change in the reported gross value of fixed assets. The nominal amount of gross investment each year is deflated to the base-year price, set as the price in 1990, using the capital goods price index.
- To infer the initial capital stock k_0 , I assume a linear depreciation process and a depreciable life of 30 years for real estate and 15 years for non-real-estate capital⁸, and then use the ratio of the accumulated depreciation over the original value to estimate the average age of capital assets and thus the "average" purchase year of the assets. I also assume the purchase year of the assets is no earlier than 1991 (or the price changes before 1991 are negligible), and infer k_0 by deflating the reported book value of fixed assets in the purchase year.
- I use the sample median of the in-use depreciation rate during the sample period, which is 0.1253 for non-real-estate assets and 0.0476 for real estate assets. The estimated depreciation rate for the overall capital is 0.0833, and this is comparable to Yan (2012) and Brandt et al. (2012). Yan (2012) finds a median depreciation rate of 9.5% in a sample of above-scale Chinese manufacturing firms during 2005-2007, and she uses 10% as the depreciation rate in constructing capital stock. Brandt et al. (2012) uses a depreciation rate of 9% in constructing capital stock with a sample of above-scale manufacturing firms during 1998-2007.
- The capital stock is imputed by the capital accumulation process $k_{t+1} = k_t(1 - \delta) + i_t$.

Table A1 reports descriptive statistics for book-value capital measures. The book-value and PIM-based investment rates are broadly consistent: the median book-value investment rate

⁸The depreciable life is estimated as (original value - salvage value) / annual depreciation for each firm-year observation assuming zero salvage value. I take the sample median. This is comparable to Wu et al. (2015); they assume 25 years of depreciable life for buildings and 40 years of depreciable life for land.

is 0.19 for non-real-estate capital and 0.09 for real estate capital, close to the PIM-based medians of 0.19 and 0.07 in Table 1. The scale of book-value capital stocks is also similar.

Table A1: Book-Value Descriptive Statistics

	Mean	SD	Min	P25	Median	P75	Max	N
log(book fixed assets, net)	20.36	1.34	17.36	19.40	20.22	21.18	24.44	19,105
log(book RE fixed assets, net)	19.59	1.27	16.45	18.70	19.50	20.35	23.38	17,217
log(book non-RE fixed assets, net)	19.63	1.57	16.15	18.48	19.50	20.65	24.10	17,218
log(book fixed assets, original)	20.79	1.35	17.85	19.82	20.64	21.62	24.93	19,574
log(book RE fixed assets, original)	19.80	1.26	16.82	18.92	19.71	20.57	23.61	19,561
log(book non-RE fixed assets, original)	20.21	1.49	16.90	19.12	20.10	21.21	24.56	19,564
Investment rate: all capital (book value)	0.30	0.49	-0.36	0.04	0.15	0.36	4.22	16,992
Investment rate: non-RE capital (book value)	0.34	0.55	-0.51	0.06	0.19	0.42	5.13	15,125
Investment rate: RE capital (book value)	0.28	0.58	-0.40	0.01	0.09	0.30	5.19	15,124

Note: Book-value capital rows are natural logs. Book-value investment rates use changes in original book value over lagged net book value.

The book value of fixed assets may understate the capital inputs, as capital can be leased without being recorded as the firm's fixed assets. Information on leased capital is only available from 2021 onwards. For those firms that appear in the sample analysed in this paper, I compute the ratio of leased capital to net fixed assets for 2021 and 2022 and find that the distribution of the ratio is highly right-skewed, with 85% of the observations below 0.1.

Another concern with the reported value of real estate assets is that land cannot be privately owned, and some land-use rights are granted by the government to firms without land-transfer fees, in which case the land is classified as intangible assets and reported at fair value. I only include land classified as fixed assets as real estate capital primarily for two reasons. First, detailed classification information on intangible assets is only available after 2007 and has many missing values, which poses a significant challenge to correct measurement. Second, land-use rights granted without fees⁹ do not involve investment payments and cannot be used as collateral. Including these assets in real estate capital would introduce further measurement error since the model accounts for costly real estate investments and pledgeable assets. As a robustness check, I also compare the data moments computed using the original sample and the sample that takes into account land in intangibles and find that although the distribution of the h/k ratio shifts to the right due to the addition of land, the other targeted data moments remain similar, and the key empirical pattern of the negative correlation between capital size and h/k ratio remains the same.

⁹ 划拨用地.

A.2 Sample Periods

The baseline descriptive sample covers 2003–2019, the longest period over which detailed fixed-asset classifications allow the construction of real estate and non-real-estate capital. I also report descriptive statistics by subperiod in Table A2. The split separates four relevant regimes: 2003–2006, before the 2007 accounting-standard reform; 2007–2010, covering the global financial crisis and the 2009–2010 four-trillion-yuan stimulus (Cong et al., 2019); 2011–2014, the early post-stimulus period overlapping with housing purchase restrictions; and 2015–2019, the later post-reform period.

The 2007 accounting-standard reform affects the interpretation of measured real estate capital because investment properties are separately classified outside fixed assets after the reform. The decline in the median RE/Non-RE ratio after 2007 is therefore partly mechanical: the median ratio falls from 0.52 in 2003–2006 to 0.44 in 2007–2010 and remains near 0.42 in the later periods. Despite this level shift, the qualitative patterns used in the paper are stable. The mean RE/Non-RE ratio remains above the median in every subperiod, and non-real-estate investment rates remain higher than real estate investment rates.

Table A2: Descriptive Statistics by Subperiod

	2003–2006	2007–2010	2011–2014	2015–2019
$\log(k)$, PIM	19.10 [19.03]	19.20 [19.07]	19.27 [19.14]	19.42 [19.29]
$\log(h)$, PIM	18.45 [18.41]	18.40 [18.35]	18.43 [18.31]	18.57 [18.48]
h/k	0.78 [0.52]	0.68 [0.44]	0.62 [0.42]	0.63 [0.42]
Investment rate: all capital	0.27 [0.13]	0.25 [0.12]	0.33 [0.17]	0.25 [0.12]
Investment rate: k	0.33 [0.15]	0.34 [0.16]	0.40 [0.23]	0.33 [0.18]
Investment rate: h	0.19 [0.08]	0.20 [0.06]	0.31 [0.11]	0.21 [0.06]
$\log(\text{EBITDA}/P_k)$	18.53 [18.44]	18.92 [18.80]	19.00 [18.85]	19.37 [19.22]
$\log(\text{total assets})$	21.31 [21.21]	21.62 [21.46]	21.87 [21.69]	22.13 [22.00]
Firm age	9.13 [9.00]	11.61 [11.00]	14.08 [14.00]	17.87 [18.00]
Firm-year observations	2,645	3,375	5,165	8,812
Firms	745	1,114	1,443	2,027

Note: Each cell reports the mean with the median in brackets. The variables use the same construction and trimming rules as Table 1. The four subperiods separate the pre-2007 accounting regime, the global financial crisis and 2009–2010 stimulus period, the early post-stimulus period overlapping with housing purchase restrictions, and the later post-reform period.

A.3 Demeaned Investment-Rate Distributions

Figure A1 reports demeaned investment-rate distributions for both assets. The top row subtracts each firm's mean investment rate across years, addressing persistent firm-level heterogeneity. The bottom row residualises investment rates on year fixed effects, addressing aggregate year shocks. In both cases the distributions shift or recenter, but the long right tail remains visible for both non-real-estate and real-estate investment rates.

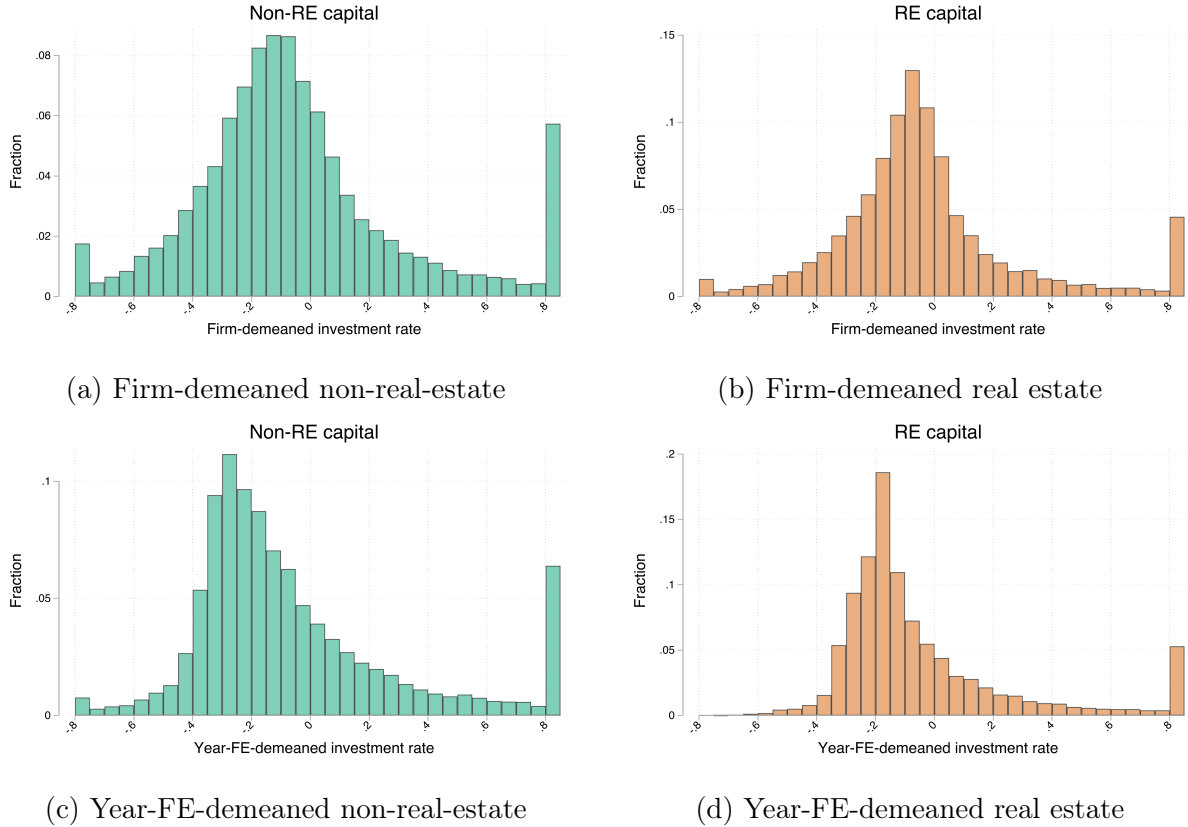


Figure A1: Demeaned Distributions of Investment Rates

Note: The figure shows the distribution of demeaned investment rates. The top row subtracts each firm's mean investment rate within the sample period. The bottom row residualises investment rates on year fixed effects. Mass outside the plotting range is clipped into endpoint bins, and spike-threshold lines are omitted because zero has a different interpretation on the demeaned scale. Source: Financial Reports of Listed Companies in China. Based on asset types classified by the author.

A.4 Sector-Specific Asset Composition

Figures A3 and A2 report the same asset-composition patterns within four manufacturing sectors: Food Manufacturing, Textiles, Nonferrous Metals, and Automobile Manufacturing.

The sector-specific size figure groups firms by full-sample quartiles of total capital. The qualitative patterns remain visible within these sectors: h/k is right-skewed, and smaller firms tend to have higher real-estate intensity than larger firms.

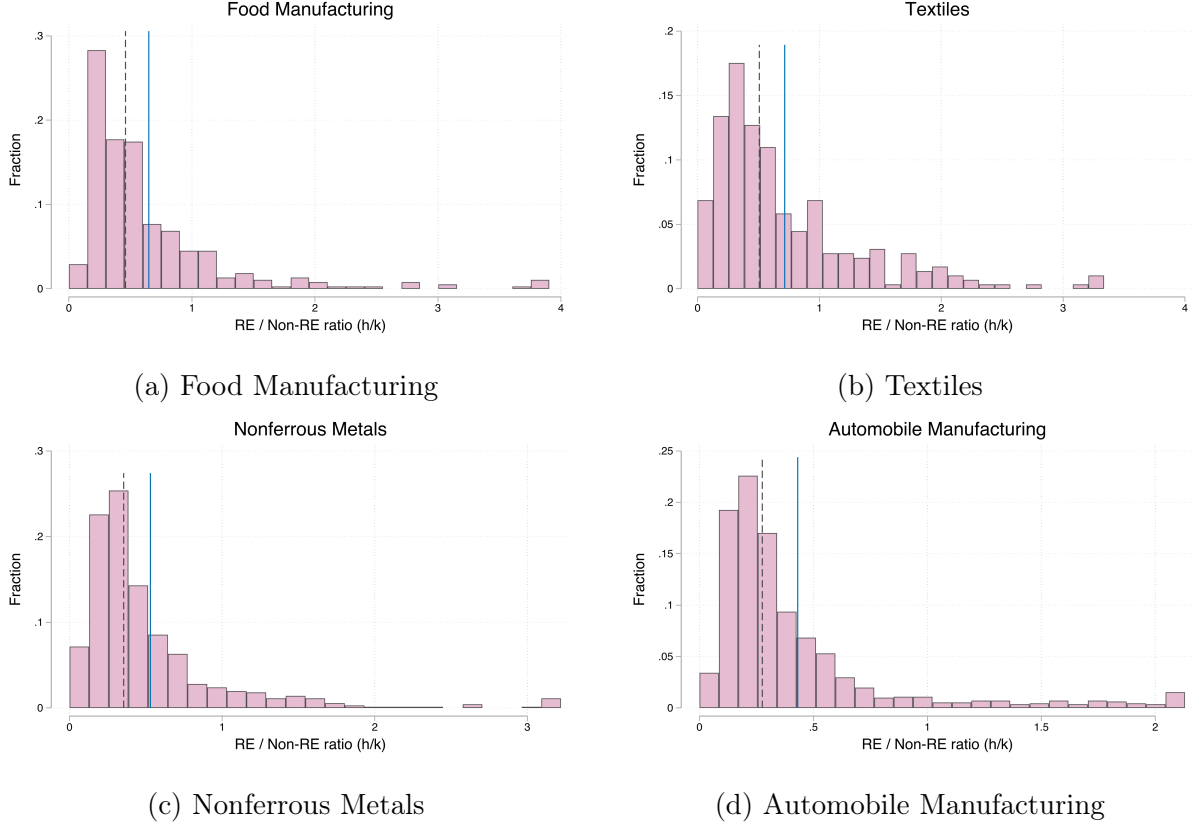


Figure A2: Sector-Specific Distributions of the RE/Non-RE Ratio

Note: Each panel reports the raw, trimmed PIM-based RE/Non-RE ratio in the 2003–2019 sample for one manufacturing sector. Display mass above the sector-specific 99th percentile is clipped into the final bin; vertical lines mark the sector mean and median. Source: Financial Reports of Listed Companies in China. Based on asset types classified by the author.

A.5 Asset Composition and Financing Variables

Figure A4 relates the RE/Non-RE capital ratio to firm size and financing variables. The negative relationship between h/k and capital size remains after adding industry fixed effects, suggesting that the size pattern documented in the main text is not driven only by sector composition. The figure also shows a positive relationship between h/k and loan ratios, but not between h/k and interest rates. This supports modelling collateral as a constraint on borrowing quantities rather than as a direct shifter of borrowing rates.

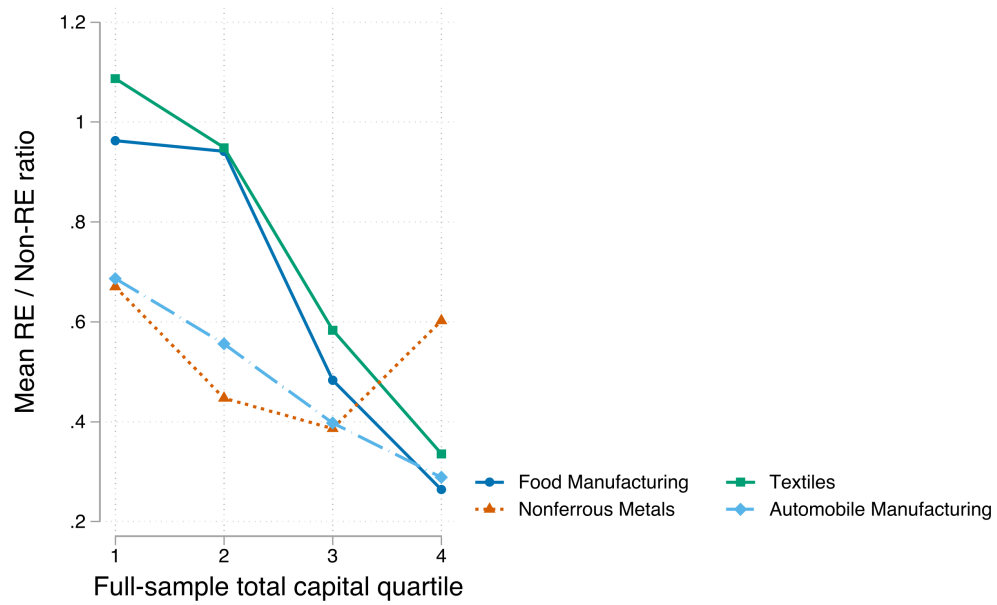


Figure A3: RE/Non-RE Ratio and Capital Size by Sector

Note: The figure plots the mean RE/Non-RE ratio by full-sample quartiles of total capital within selected manufacturing sectors. Source: Financial Reports of Listed Companies in China. Based on asset types classified by the author.

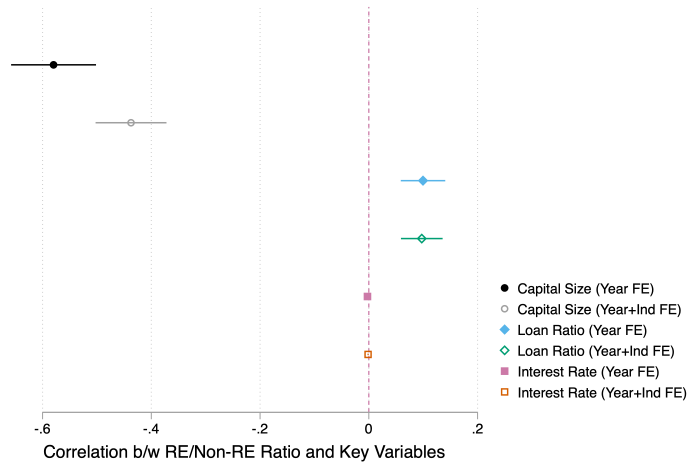


Figure A4: RE/Non-RE Ratio and Financing Variables

Note: This graph displays estimated coefficients for the relationship between RE/Non-RE ratio and key variables in Chinese listed manufacturing firms (2003-2019). RE/Non-RE ratio represents the ratio of real estate to non-real-estate capital stock constructed using the perpetual inventory method. Size measure is $\log(\text{RE} + \text{Non-RE capital})$, debt ratio is $(\text{short-term} + \text{long-term loans}) / \text{net fixed assets}$, and interest rate is $\text{financial expenses} / \text{total debt}$. All relationships are estimated with standard errors clustered at the firm level, controlling for firm size ($\log$ total capital) when measuring the debt ratio or interest rate. Two specifications are shown for each outcome: with year fixed effects only (filled markers) and with both year and industry fixed effects (hollow markers). Confidence intervals are at the 95% level. Source: Financial Reports of Listed Companies in China. Based on asset types classified by the author.

A.6 Relative Price of Real Estate Capital

Empirically, I obtain the relative price of real estate capital p as follows:

- For each firm-year observation, I compute the ratio of nominal value (sum of past purchases - depreciation) to the real value of real estate capital (constructed by PIM). This can be considered as a weighted average of real estate capital purchase prices in the years before time t , weighted by the amount of real estate capital for each firm.
- I normalise the prices from the previous step by the price of non-real-estate capital so that the prices are relative to the price of k . The 1% and 99% percentiles of the obtained prices are 0.78 and 4.62. The median is 1.56, and the mean is 1.76.

p is set as the sample mean of these prices. As a comparison, the relative price of buildings and structures to machines from the Price Index of Capital Goods ranges from 1.03 to 3.03 from 1990 to 2019. The mean and median of this relative price series are 1.93 and 1.896, respectively.

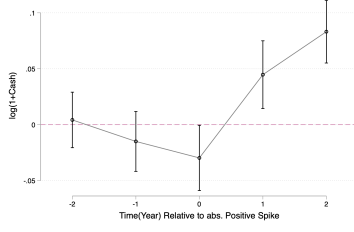
B Cash/Debt and Investment Spikes

This section provides supporting evidence for the interpretation of the collateral constraint in Section 4. It examines whether firms finance lumpy investment episodes mainly through cash accumulation or debt.

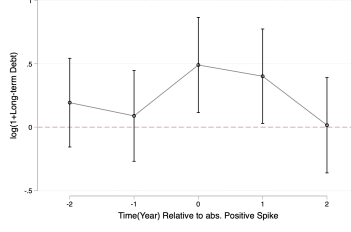
Görtz et al. (2023) shows that there is a build-up of cash stock before (anticipated) positive investment spikes. The cash stock drops during the lumpy adjustment period and recovers afterwards. This indicates that investments are primarily financed through internal funding. Following this reasoning, I test the dynamics of cash and debt around investment spikes by running the following regression:

$$d_{it} = \sum_{h=-2}^2 \beta_h \times 1\{spike_{i,t+h}\} + \delta_t + \mu_i + x_{it} + \varepsilon_{it},$$

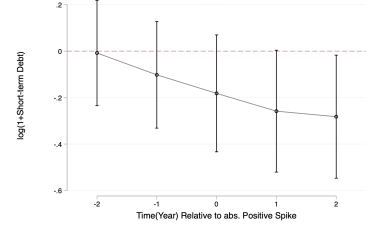
where the left-hand side variable d_{it} is the logged book value of cash, defined as cash and liquid financial assets, short-term debt, or long-term debt. Investment spikes are defined as absolute spikes when investment rates are greater than 0.3 or smaller than -0.3. Note that investment spikes can last for more than one period. The sample median and mean duration of positive investment spikes are 2 and 2.76 periods, respectively. β_h measures the average value of d two periods before and after the spike events, after removing the year and firm fixed effects and controlling for x (size and ownership structure).



(a) Cash and Cash Equivalent

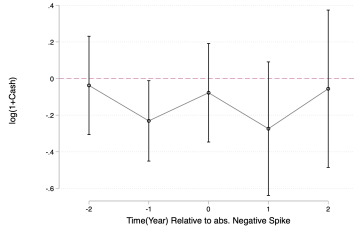


(b) Long-Term Debt

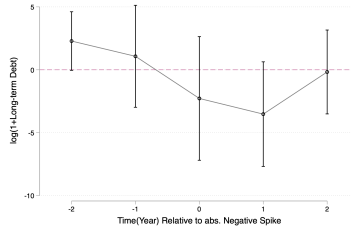


(c) Short-term Debt

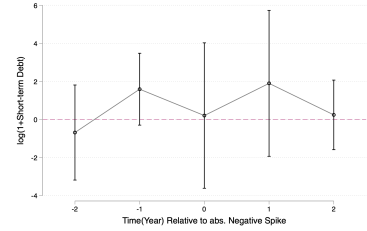
Figure A5: Cash/Debt Dynamics around Positive Investment Spikes



(a) Cash and Cash Equivalent



(b) Long-Term Debt



(c) Short-term Debt

Figure A6: Cash/Debt Dynamics around Negative Investment Spikes

Figure A5 shows the cash and debt dynamics around positive investment spikes. Panel (a) indicates that there is no significant cash build-up before the investment spikes. There is a slight decrease in cash during the lumpy adjustment and an increase in cash after the spike. Panels (b) and (c) show that there is no significant change in short-term debt (due in less than one year), but a significant increase in long-term debt during the spike and one year after, followed by a significant drop two years after the spike. These results suggest that lumpy investments are financed more through debt rather than cash, and that the debt is repaid relatively quickly (within two years). Figure A6 shows no significant fluctuations in cash and debt around negative investment spikes. I also perform the test for relative investment spikes, and the results are similar.

C Revenue Function Estimation

This section reports the complete revenue-function estimates used to pin down the production technology and profitability process before the SMM estimation.

Table A3: Quasi-differenced Revenue Function Estimation

	$\hat{\rho}_z$	$\hat{\alpha}$	$\hat{a}$	$Dist$
$\sigma = 0.05$	0.7077 (0.011)	0.7659 (0.012)	0.6857 (0.012)	0.1566
$\sigma = 0.1526$	0.7082 (0.011)	0.7587 (0.012)	0.6694 (0.011)	0.1604
$\sigma = 0.2553$	0.7100 (0.011)	0.7581 (0.012)	0.6501 (0.011)	0.1635
$\sigma = 0.3579$	0.7127 (0.011)	0.7561 (0.012)	0.6307 (0.012)	0.1674
$\sigma = 0.4605$	0.7163 (0.011)	0.7487 (0.012)	0.6132 (0.012)	0.1724
$\sigma = 0.5632$	0.7203 (0.011)	0.7434 (0.012)	0.5947 (0.014)	0.1761
$\sigma = 0.6658$	0.7241 (0.012)	0.7389 (0.012)	0.5759 (0.015)	0.1789
$\sigma = 0.7684$	0.7275 (0.012)	0.7350 (0.012)	0.5569 (0.017)	0.1811
$\sigma = 0.8711$	0.7306 (0.012)	0.7318 (0.012)	0.5377 (0.019)	0.1828
$\sigma = 0.9737$	0.7332 (0.012)	0.7289 (0.012)	0.5184 (0.021)	0.1841
$\sigma = 1.0763$	0.7356 (0.012)	0.7265 (0.012)	0.4992 (0.024)	0.1852
$\sigma = 1.1789$	0.7377 (0.012)	0.7244 (0.012)	0.4802 (0.026)	0.1861
$\sigma = 1.2816$	0.7395 (0.012)	0.7225 (0.013)	0.4613 (0.028)	0.1869
$\sigma = 1.3842$	0.7411 (0.012)	0.7208 (0.013)	0.4426 (0.030)	0.1876
$\sigma = 1.4868$	0.7426 (0.012)	0.7192 (0.013)	0.4241 (0.033)	0.1882
$\sigma = 1.5895$	0.7440 (0.012)	0.7178 (0.013)	0.4061 (0.035)	0.1888
$\sigma = 1.6921$	0.7452 (0.012)	0.7165 (0.013)	0.3883 (0.037)	0.1892
$\sigma = 1.7947$	0.7464 (0.012)	0.7153 (0.013)	0.3709 (0.038)	0.1897
$\sigma = 1.8974$	0.9000 (0.008)	0.4085 (0.021)	0.7298 (0.076)	0.2014
$\sigma = 2$	0.9000 (0.008)	0.4092 (0.021)	0.7318 (0.080)	0.2014

Table A4: Quasi-differenced Revenue Function Estimation: Balanced Panel

	$\hat{\rho}_z$	$\hat{\alpha}$	$\hat{a}$	$\hat{\sigma}$	$Dist$
$\sigma = 0.05$	0.8469 (0.015)	0.6497 (0.032)	0.7255 (0.026)	n.a.	83.55
$\sigma = 0.5375$	0.8495 (0.015)	0.6352 (0.031)	0.6471 (0.030)	n.a.	87.22
$\sigma = 1.0250$	0.8497 (0.015)	0.6350 (0.031)	0.5744 (0.052)	n.a.	87.16
$\sigma = 1.5125$	0.8502 (0.015)	0.6342 (0.031)	0.5082 (0.079)	n.a.	86.96
$\sigma = 2.00$	0.8505 (0.015)	0.6335 (0.031)	0.4448 (0.105)	n.a.	86.86

Note: The estimates are obtained with a two-step GMM estimator. The sample is a balanced panel with 785 firms and 5 years (2015-2019).

D Simulated Method of Moments

This section describes the model solution, global optimisation routine, and post-estimation calculations used in the SMM exercise.

D.1 Model Solution

The model is solved by value function iteration. There are dynamic choices, on both extensive and intensive margins, of two capital inputs. The value function iteration follows the following steps:

1. z is discretised with the Tauchen method on a coarse grid, denoted as zz . k and h are discretised on coarse grids (kk, hh) and fine grids $(\widetilde{kk}, \widetilde{hh})$.
2. The discretised value function V defined in equation 2 is on $(zz, \widetilde{kk}, \widetilde{hh})$. The discretised value functions V^a, V^k, V^h, V^n defined in equation 3, 4, 5, and 6 are on (zz, kk, hh) . $\widetilde{kk}$ are such that for any $k \in kk$, $k(1 - \delta_k) \in \widetilde{kk}$. The same applies to $\widetilde{hh}$.
3. For a guess $V_i(zz, \widetilde{kk}, \widetilde{hh})$:
 - For each $V_{i+1}^a, V_{i+1}^k, V_{i+1}^h, V_{i+1}^n$, find the optimal $(k, h) \in (\widetilde{kk}, \widetilde{hh})$ that maximises the right-hand side for every (z, k, h) state and update the values on the intensive margin.

- Evaluate $V_{i+1}^a, V_{i+1}^k, V_{i+1}^h, V_{i+1}^n$ on the fine grid $(zz, \widetilde{kk}, \widetilde{hh})$, find the maximum among the four cases, and update the value and policy on the extensive margin. This gives $V_{i+1}(zz, \widetilde{kk}, \widetilde{hh})$
4. Given the converged V and V^a, V^k, V^h, V^n :
- Construct new grids that are finer for (z, k, h) , denoted as $(\widehat{zz}, \widehat{kk}, \widehat{hh})$. $\widehat{kk}$ are such that for any $k \in \widehat{kk}$, $k(1 - \delta_k) \in \widehat{kk}$. The same applies to $\widehat{hh}$.
 - Interpolate V on $(\widehat{zz}, \widehat{kk}, \widehat{hh})$. For each state $(z, k, h) \in (\widehat{zz}, \widehat{kk}, \widehat{hh})$, find the policies on the extensive margin and intensive margin. This is to allow for simulation on the discretised state space. The simulation on the fine $\widehat{zz}$ ensures the variation in z , and the simulation on $\widehat{kk}, \widehat{hh}$ avoids the errors from interpolating around the kinks.

D.2 Optimisation Algorithm

The algorithm is designed to find the global optimum by combining quasi-random initial sampling with iterative local optimisation. It consists of two main stages: the initial guess generation and the global optimisation search.

1. Initial Guess Generation

The algorithm begins by generating initial guesses for the parameters θ using a Halton sequence, which is a low-discrepancy, quasi-random sequence. This sequence is ideal for sampling because it covers the parameter space uniformly, reducing the risk of missing important regions.

- (a) **Parameter Space Definition:** Let θ be a vector of d parameters, each with bounds $\theta_i \in [l_i, u_i]$ for $i = 1, 2, \dots, d$.
- (b) **Halton Sampling:** The algorithm generates N points $\theta^{(1)}, \theta^{(2)}, \dots, \theta^{(N)}$ using the Halton sequence, ensuring that these points are uniformly distributed across the parameter space. Each point $\theta^{(i)}$ is scaled to respect the bounds $[l_i, u_i]$.
- (c) **Targeted Perturbation:** For a subset of parameters $\theta_{\text{perturb}} \subset \theta$, the algorithm optionally adds normally distributed perturbations to create a denser sampling in key areas. This generates additional points near each $\theta^{(i)}$, enhancing the sampling density in critical regions.

2. Global Optimisation Search

- (a) **Setup and Initialisation:** The search begins by initialising storage for local optima and setting up the bounds l_i and u_i for each parameter.

(b) **Iterative Local Search:** The algorithm iteratively refines its search for the global minimum by:

- i. **Initial Guess Selection:** For each iteration i , the starting point $\theta^{(i)}$ is selected from the set of best initial guesses generated by the Halton sequence. In the first iteration, $\theta^{(1)}$ is directly taken from the Halton points. For subsequent iterations, $\theta^{(i)}$ is computed as a weighted combination:

$$\theta^{(i)} = (1 - w_i)\theta_{\text{halton}}^{(i)} + w_i\theta_{\text{local}}^{(i-1)},$$

where $\theta_{\text{halton}}^{(i)}$ is the next Halton point, $\theta_{\text{local}}^{(i-1)}$ is the best local minimum from previous iterations, and w_i is a weight that increases with each iteration.

- ii. **Local Optimisation:** The Nelder-Mead Subplex algorithm is applied starting from $\theta^{(i)}$ to find a local minimum $\theta_{\text{local}}^{(i)}$ and its corresponding objective function value $J(\theta_{\text{local}}^{(i)})$. This local minimum is then saved.

(c) **Convergence Check:** The algorithm checks for convergence by evaluating the difference ϵ between consecutive local minima:

$$\epsilon = \|\theta_{\text{local}}^{(i)} - \theta_{\text{local}}^{(i-1)}\|_{\infty},$$

where $\|\cdot\|_{\infty}$ denotes the maximum norm. If $\epsilon \leq \text{tol}$, the algorithm converges; otherwise, it continues to the next iteration. The process also stops if the maximum number of iterations N^* is reached.

(d) **Result Compilation:** After the iterations, the algorithm identifies the global minimum θ^* by selecting the parameter set with the lowest objective function value $J(\theta^*)$ from all local minima found. This global minimum is then saved and returned.

D.3 Standard Errors and Elasticity Matrix

This section outlines the methodology used to compute the standard errors of parameter estimates and the elasticity matrix as part of the post-estimation analysis. The process leverages local polynomial regression to approximate gradients and derive the Jacobian matrix.

1. Simulate moments around the parameter estimates: To calculate how the moments change with respect to the parameters, I generate a set of perturbed parameter values around the estimate $\hat{x}$. These values form a symmetric neighbourhood around $\hat{x}$. I simulate the corresponding moments for each $x_i \in x_{\text{vec}}$.
2. Local polynomial regression to compute derivatives: I approximate the gradient of the simulated moments with respect to the parameters using local polynomial regression. Given perturbed values x_{vec} and corresponding moment values m_{vec} , I fit a polynomial of degree $p = 3$.

3. Compute the gradients and the elasticities for each parameter x : For a model with k moments and n parameters, the Jacobian matrix J is a $k \times n$ matrix, where each element J_{ij} is

$$J_{ij} = \frac{\partial m_i}{\partial x_j},$$

and the entry of the elasticity matrix E is given by

$$E_{ij} = \frac{\partial m_i}{\partial x_j} \cdot \frac{\hat{x}_i}{m_j(\hat{x}_i)} = J_{ij} \cdot \frac{\hat{x}_j}{m_i(\hat{x}_j)}$$

4. Compute the standard errors: The variance-covariance matrix of the parameter estimates is

$$\text{VCV}_\theta = (J'WJ)^{-1}$$

where:

- J is the Jacobian matrix,
- W is the weighting matrix, which is the inverse of the variance-covariance matrix of the data moments.

The standard errors of the parameter estimates are the square roots of the diagonal elements of the variance-covariance matrix:

$$\text{SE}(\theta_j) = \sqrt{(\text{VCV}_\theta)_{jj}}$$

E Additional Estimation Results

This section reports untargeted moments and auxiliary model-fit evidence used to assess the preferred specification.

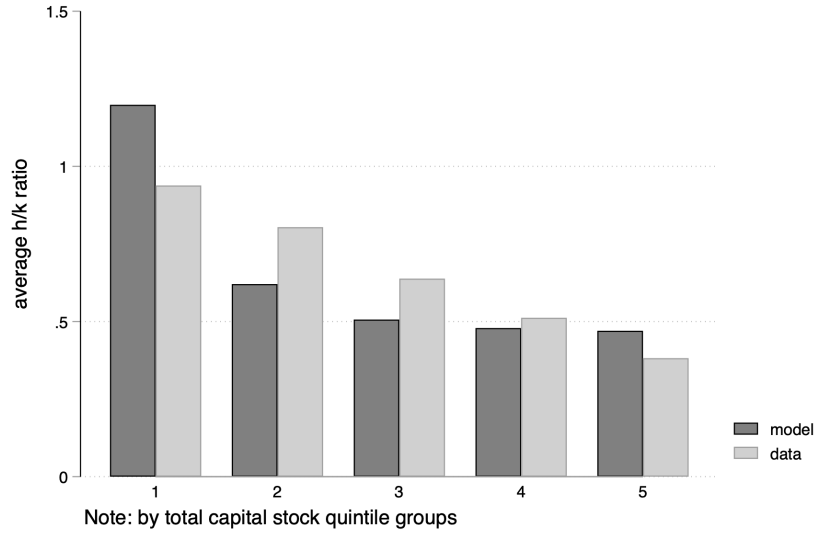
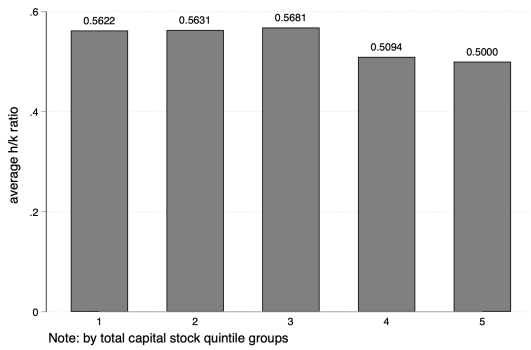
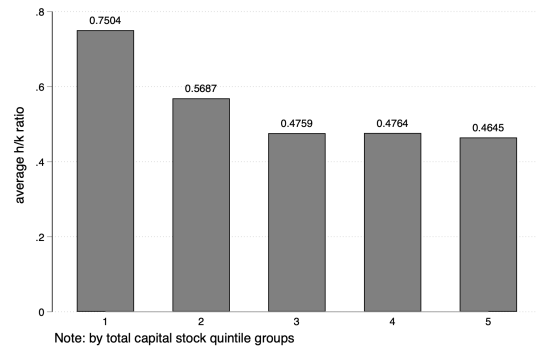
E.1 Untargeted Moments

E.2 Firm Size and Binding Collateral Constraints

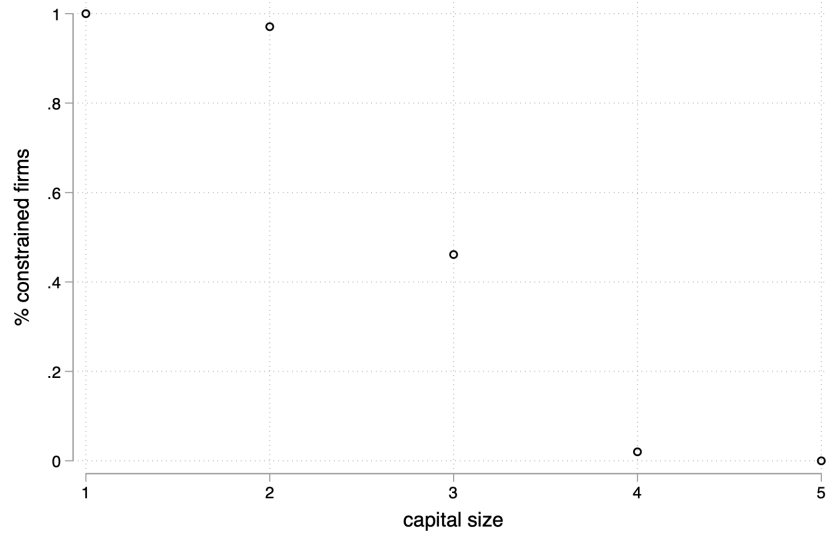
Figure A7 compares the average h/k ratio across capital-size quintile groups in the data and in simulated economies. In these simulations, γ , F_k , ω , and F_h are fixed at their preferred estimates from Table 4, column (4), while the pledgeability parameters are varied. The figure provides a visual counterpart to the model comparison in Table 4: changing the collateral-constraint environment weakens the model's ability to reproduce the declining h/k ratio across capital-size groups.

Table A5: Untargeted Moments: Relative Investment Spikes

	(1)	(2)	(3)	(4)	Data
	no cons.	internal funds	sym. pledge	asym. pledge	
Untargeted Moments					
$rel.spike_k^+$	0.18	0.13	0.19	0.19	0.11
$rel.spike_h^+$	0.14	0.10	0.14	0.16	0.14
$rel.spike_k^-$	0.00	0.00	0.00	0.00	0.01
$rel.spike_h^-$	0.04	0.00	0.04	0.04	0.02

(a) $\phi_k > 0, \phi_h > 0$ (b) $\phi_k = \phi_h = 0$ (c) $\phi_k = \phi_h = +\infty$ Figure A7: h/k ratio and Capital Size in Simulation

Note: In the simulation, γ, F_k, ω, F_h are the estimates as in Table 4 column (4). Total capital stock is defined as $k+h$. The observations are grouped into five bins according to the quintiles of total capital stock. The graph plots the average h/k ratio for each group.



Note: A firm is constrained if its constrained value is less than 97% of its unconstrained value. The constrained-value comparison captures the effect of potentially binding future constraints, rather than only current-period corner investment choices.

Figure A8: Constrained firms and Capital Size

Figure A8 shows the share of financially constrained firms by capital-size quintile in the preferred model simulation. A firm is defined as constrained if its constrained value is less than 97% of its unconstrained value. Smaller firms are more likely to be constrained, and the constrained share declines with capital size. This pattern coincides with the declining h/k ratio across size groups in the preferred simulation.